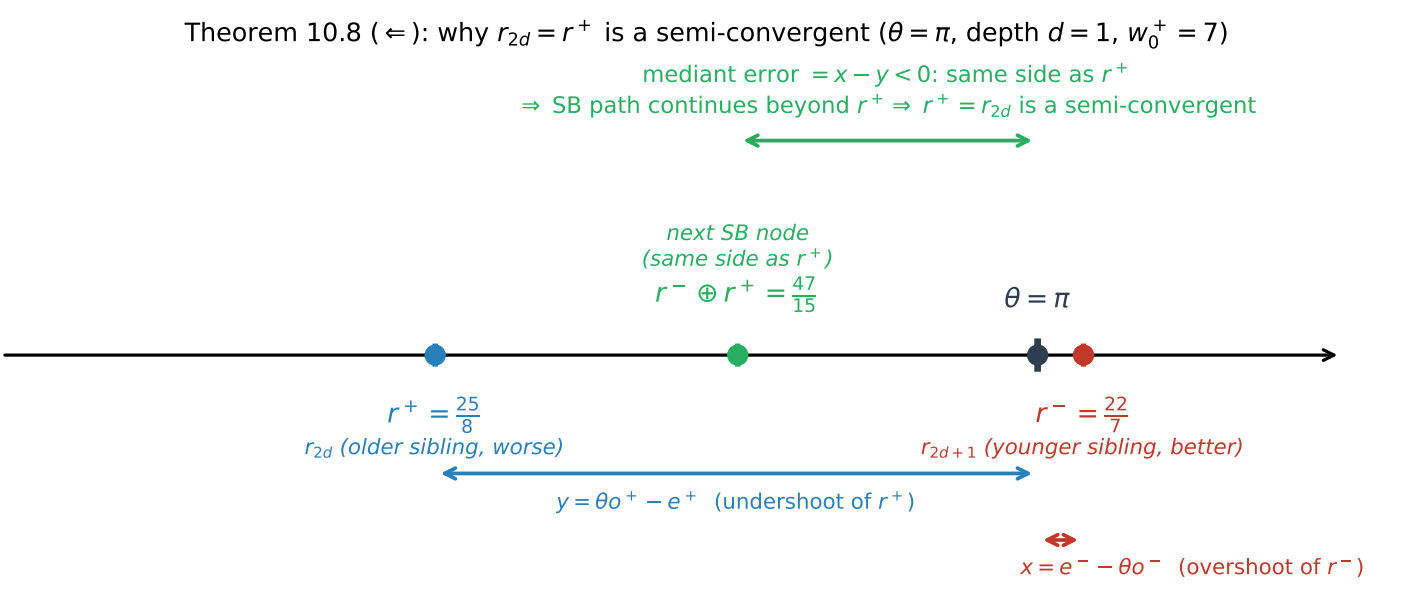
Theorem 10.8 (⇐): why $r_{2d} = r^+$ is a semi-convergent ($\theta = \pi$, depth $d = 1$, $w_0^+ = 7$)
mediant error $= x - y < 0$: same side as r^+
⇒ SB path continues beyond r^+ ⇒ $r^+ = r_{2d}$ is a semi-convergent
next SB node
(same side as r^+)
$r^- \oplus r^+ = \frac{47}{15}$
$\theta = \pi$
$r^+ = \frac{25}{8}$
r_{2d} (older sibling, worse)
$r^- = \frac{22}{7}$
r_{2d+1} (younger sibling, better)
$y = \theta o^+ - e^+$ (undershoot of r^+)
$x = e^- - \theta o^-$ (overshoot of r^-)